

MEDVEZHYEGORSK, Russian Federation

WMO: 227210

Lat: **62.917N**Lon: **34.433E**Elev: **81**StdP: **100.36**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.8	-26.5	-32.5	0.2	-29.7	-29.0	0.3	-26.5	4.5	-5.6	4.2	-5.4	1.0	290	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	26.6	19.0	24.7	18.1	22.8	17.0	20.4	24.7	19.2	23.1	17.9	21.4	1.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.7	22.6	17.5	12.7	21.4	16.3	11.7	20.1	58.9	24.8	54.9	23.2	50.9	21.4	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.2	3.5	3.2	DB	-31.5	29.0	4.4	2.3	-34.6	30.7	-37.2	32.0	-39.6	33.3	-42.8	35.0	(4)
(5)			WB	-31.6	22.1	4.2	1.9	-34.6	23.5	-37.1	24.6	-39.5	25.7	-42.6	27.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-10.2	-10.3	-5.1	1.4	7.5	13.4	16.9	14.5	9.4	3.3	-2.5	-6.6	(6)
(7)		DBStd	10.69	8.01	7.95	5.50	4.08	4.68	4.19	3.55	3.29	3.43	4.02	5.16	7.01	(7)
(8)		HDD10.0	3191	626	567	469	259	105	13	1	3	51	208	375	514	(8)
(9)		HDD18.3	5737	884	801	727	508	336	157	69	124	267	465	625	772	(9)
(10)		CDD10.0	538	0	0	0	1	29	116	214	144	33	1	0	0	(10)
(11)		CDD18.3	42	0	0	0	0	1	9	25	6	0	0	0	0	(11)
(12)		CDH23.3	315	0	0	0	0	16	86	172	42	0	0	0	0	(12)
(13)		CDH26.7	48	0	0	0	0	2	17	26	3	0	0	0	0	(13)
(14)	Wind	WSAvg	1.4	1.4	1.4	1.5	1.3	1.4	1.3	1.2	1.3	1.4	1.4	1.5	(14)	
(15)	Precipitation	PrecAvg	716	55	42	40	35	53	66	87	80	68	68	63	59	(15)
(16)		PrecMax	898	93	70	82	65	116	135	217	136	186	138	94	106	(16)
(17)		PrecMin	514	22	10	12	5	18	24	35	28	16	28	8	17	(17)
(18)		PrecStd	98	17	18	14	18	22	27	41	29	32	31	20	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.8	3.3	8.9	16.2	25.5	28.9	29.2	26.8	20.5	13.2	7.7	5.4	(19)
(20)			MCWB	1.8	2.1	3.7	9.6	14.8	19.0	20.1	20.1	15.5	11.6	6.7	4.9	(20)
(21)		2%	DB	1.3	2.0	5.2	12.1	21.2	25.7	27.0	24.4	18.0	11.3	6.4	3.2	(21)
(22)			MCWB	0.8	1.1	1.6	6.9	12.9	17.6	19.9	18.1	14.1	10.3	5.8	2.6	(22)
(23)		5%	DB	0.6	0.9	3.0	9.8	18.4	23.2	25.3	22.4	16.1	10.1	5.1	2.1	(23)
(24)			MCWB	0.2	0.4	0.8	5.5	11.5	16.3	18.9	17.2	13.3	9.2	4.8	1.5	(24)
(25)		10%	DB	-0.2	0.0	1.7	7.5	15.6	20.9	23.5	20.6	14.7	9.0	3.8	1.1	(25)
(26)			MCWB	-0.5	-0.4	0.2	4.0	10.2	15.2	18.0	16.1	12.6	8.1	3.3	0.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.4	4.2	9.9	16.5	20.8	22.5	21.3	16.4	12.0	7.2	5.0	(27)
(28)			MCDWB	2.6	3.1	7.9	14.8	22.9	26.1	27.0	24.9	19.2	12.8	7.6	5.6	(28)
(29)		2%	WB	1.0	1.3	2.2	7.5	14.1	19.1	21.0	19.3	14.9	10.6	5.9	2.8	(29)
(30)			MCDWB	1.2	1.7	4.1	11.2	19.9	24.2	25.4	23.2	17.0	11.2	6.2	3.1	(30)
(31)		5%	WB	0.3	0.5	1.3	5.9	12.4	17.4	19.8	17.8	13.8	9.3	4.7	1.7	(31)
(32)			MCDWB	0.6	0.9	2.6	9.1	17.1	21.8	23.6	21.0	15.4	10.0	5.1	1.9	(32)
(33)		10%	WB	-0.5	-0.5	0.5	4.3	10.7	15.6	18.7	16.6	12.9	8.1	3.4	0.8	(33)
(34)			MCDWB	-0.2	0.0	1.6	7.0	15.0	20.0	22.2	19.7	14.4	8.9	3.8	1.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	6.9	8.1	8.5	9.4	9.3	8.7	8.4	6.8	4.4	4.1	4.8	(35)
(36)			MCDBR	4.0	4.7	7.8	12.1	14.3	12.5	11.6	11.6	9.3	5.3	4.0	3.7	(36)
(37)			MCWBR	4.1	4.4	5.5	7.4	7.0	5.9	5.4	6.0	5.8	4.3	4.0	3.7	(37)
(38)		5% WB	MCDWB	3.8	4.4	6.4	10.9	11.8	10.7	10.0	9.6	7.5	4.8	3.9	3.6	(38)
(39)			MCWBR	3.9	4.2	4.8	7.1	6.7	5.9	5.6	5.6	5.2	4.2	3.9	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.235	0.288	0.314	0.352	0.353	0.346	0.379	0.361	0.333	0.312	0.246	0.210	(40)	
(41)		taud	1.993	2.082	2.145	2.278	2.445	2.495	2.420	2.446	2.511	2.390	2.079	1.956	(41)	
(42)		Ebn at Noon	384	630	773	828	864	877	832	816	773	628	370	210	(42)	
(43)		Edh at Noon	33	69	96	105	97	95	100	89	69	52	30	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.19	0.77	2.21	3.69	4.88	5.42	5.04	3.75	2.11	0.79	0.20	0.08	(44)	
(45)		RadStd	0.03	0.17	0.31	0.37	0.45	0.39	0.58	0.35	0.23	0.11	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	-204	N/A	N/A

Nomenclature: See separate page